

cse 438 – robotics

SMART PLANT MONITORING BOT



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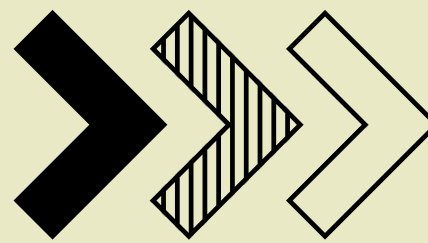
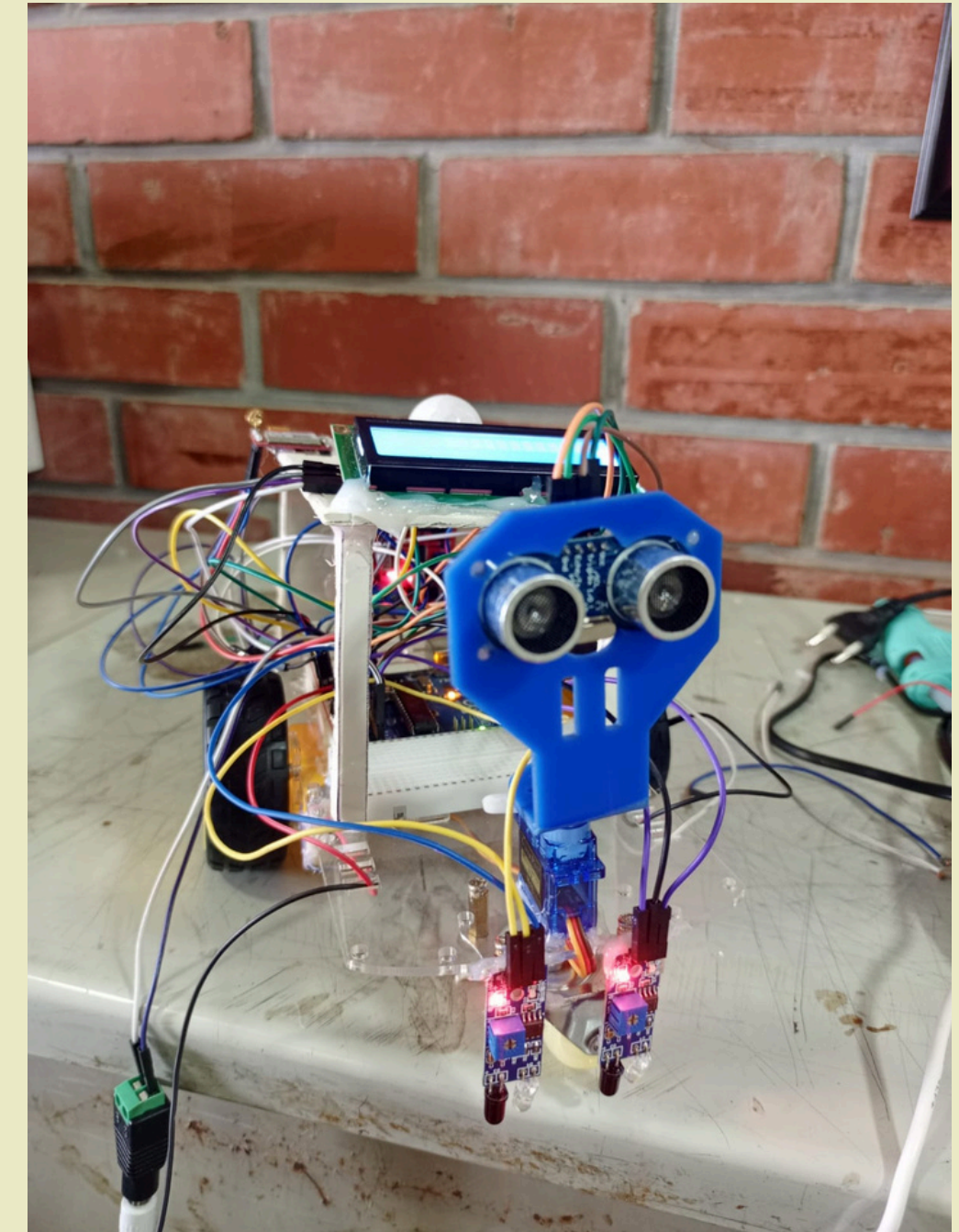


Executive Summary

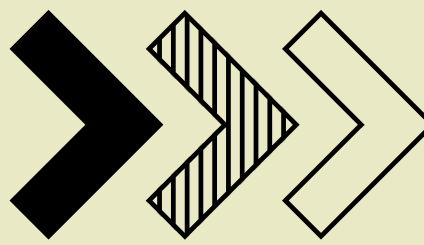
Smart Plant Monitoring Bot is an Arduino-based robot designed to monitor and protect plants automatically. It follows a black line around the plant area, detects obstacles, and gives alerts when any unwanted movement is detected near the plants.

This bot can:

- Follow a black line using IR sensors
- Patrol around the plant area automatically
- Detect obstacles using an ultrasonic sensor
- Scan left and right using a servo motor
- Find a clear path and return to the line
- Detect motion near plants using a PIR sensor
- Give buzzer alert during danger
- Show alert messages on I2C LCD display
- Send SMS alert to the owner using GSM module
- Help protect plants from cats, dogs, birds or unwanted movement



Contribution



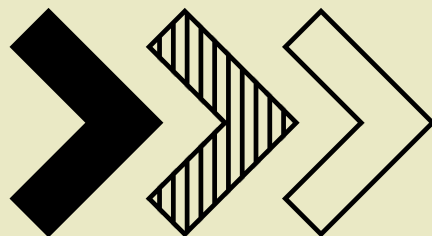
Coding

- Line-following logic using IR sensors
- Obstacle detection using ultrasonic sensor
- Servo-based path scanning
- Clear path selection logic
- PIR-based motion detection
- Buzzer, LCD, and GSM alert integration
- Sensor condition testing

Algorithm & Flow Analysis

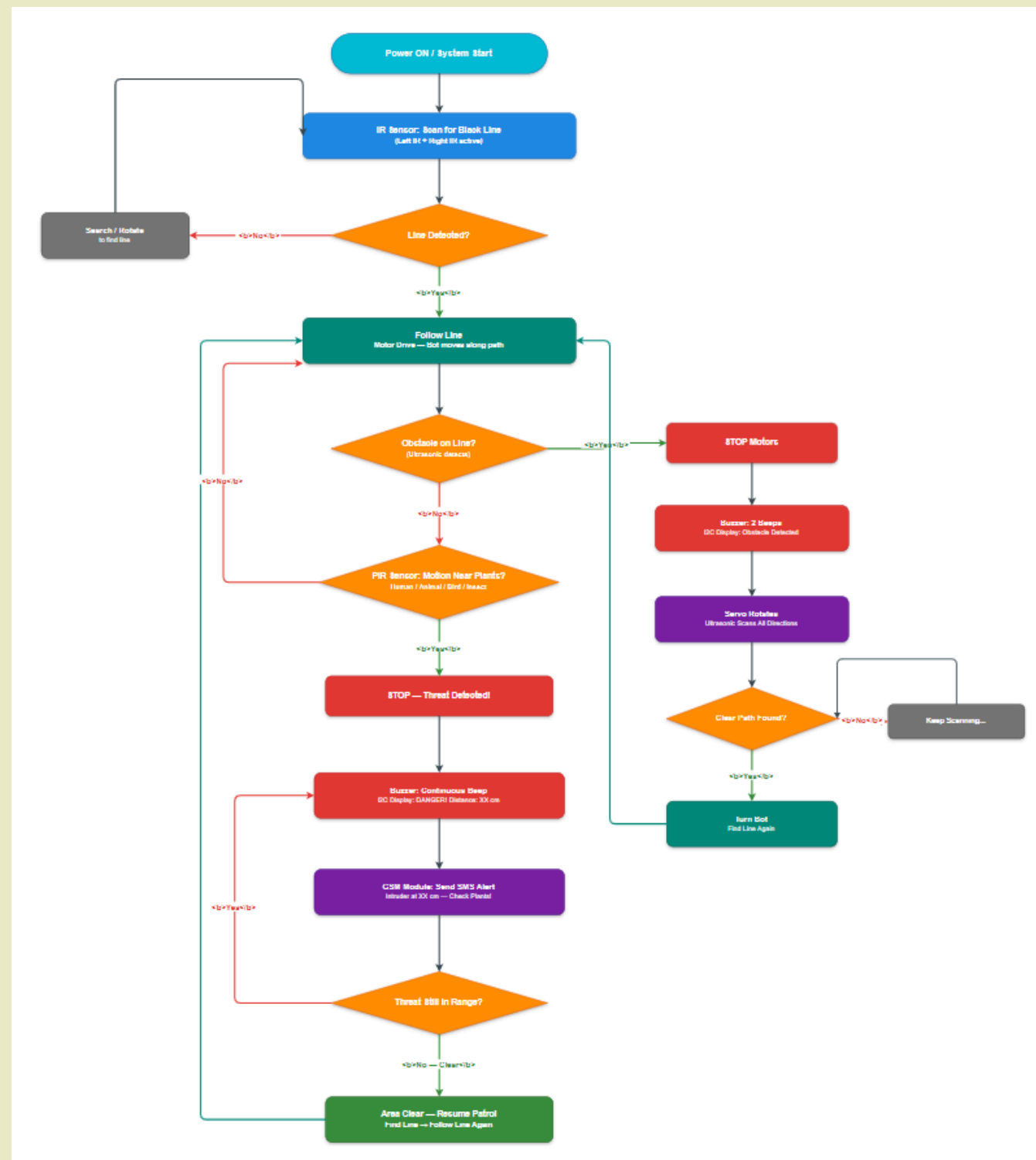
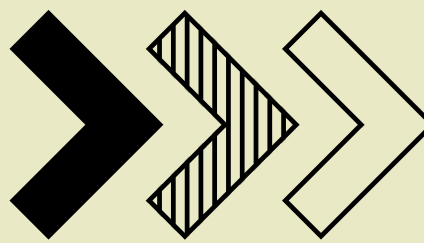
- Robot working flow design
- Line-following algorithm
- Obstacle avoidance algorithm
- Danger alert workflow
- Pseudocode preparation
- Flow diagram design
- Output testing and validation

Equipment & Components



Component	Quantity	Purpose
Arduino Uno	1	Main microcontroller for robot control
IR Sensor	2	Detects black line for line following
Ultrasonic Sensor	1	Measures distance and detects obstacles
Servo Motor	1	Rotates the ultrasonic sensor for path scanning
PIR Sensor	1	Detects motion near the plants
GSM Module	1	Sends SMS alerts to the owner
16x2 I2C LCD	1	Displays status messages and alerts
L298N Motor Driver	1	Controls the DC motors
DC Gear Motor	2	Drives the robot wheels
Buzzer	1	Provides sound alerts
Mini Chassis with Wheels	1	Supports robot movement and structure
Battery / Power Supply	1	Supplies power to the system

Flow Diagram & Pseudocode



```
START

Setup all sensors, motors, LCD, GSM
Show "System Ready" on LCD

REPEAT forever:

    Read PIR sensor
    Read distance from ultrasonic
    Read left IR and right IR

    // --- CHECK 1: PIR threat ---
    IF motion detected THEN
        Stop bot
        Send SMS with distance
        Beep continuously
        Show "DANGER" on LCD
        WAIT until threat is gone
        Stop buzzer
        Go back to line follow

    // --- CHECK 2: Obstacle ---
    ELSE IF obstacle closer than 20cm THEN
        Stop bot
        Beep 2 times
        Show "Obstacle Found" on LCD
        Scan left and right with servo
        Go to clearer side
        Search for line again
        Resume patrol

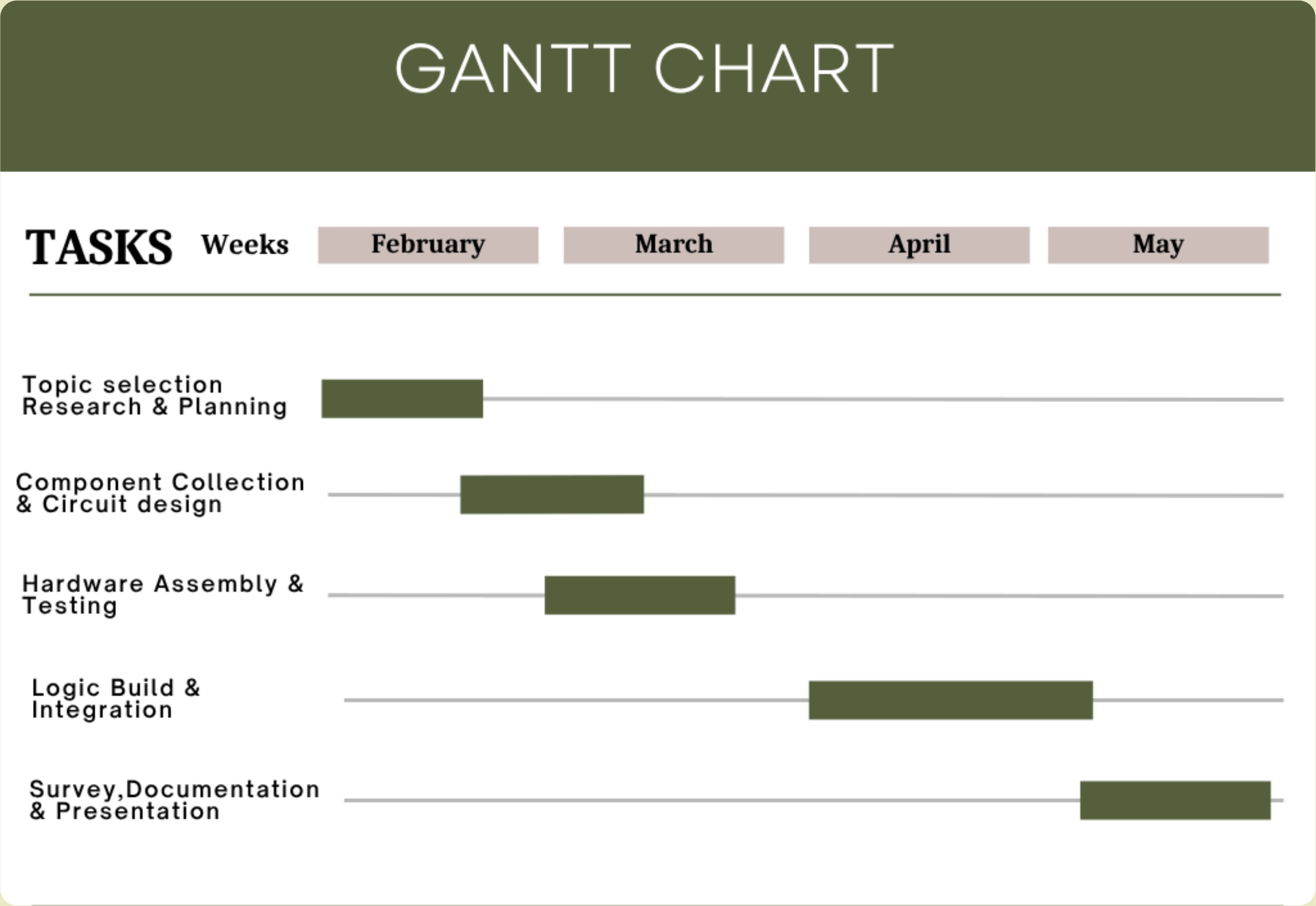
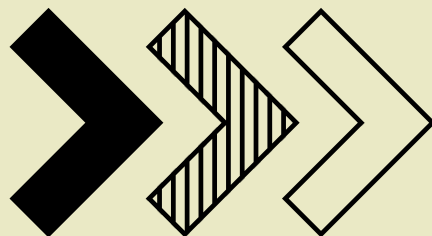
    // --- CHECK 3: Line Follow ---
    ELSE
        IF both IR on black line → go forward
        IF only left IR on black → turn left
        IF only right IR on black → turn right
        IF both IR off line → search line

    END IF

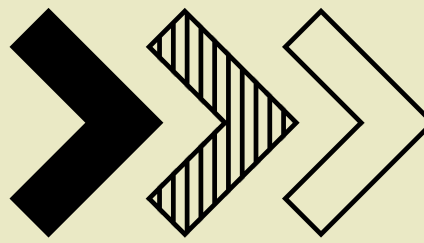
END REPEAT

STOP
```


Gant Chart



Code Snippets



Motor Control Logic

```
void forward() {
  analogWrite(enA, MOTOR_SPEED);
  analogWrite(enB, MOTOR_SPEED);

  digitalWrite(in1, LOW);
  digitalWrite(in2, HIGH);

  digitalWrite(in3, HIGH);
  digitalWrite(in4, LOW);
}

void Stop() {
  analogWrite(enA, 0);
  analogWrite(enB, 0);

  digitalWrite(in1, LOW);
  digitalWrite(in2, LOW);
  digitalWrite(in3, LOW);
  digitalWrite(in4, LOW);
}
```

IR-Based Line Following Logic

```
void lineFollow() {
  int leftIR = digitalRead(L_S);
  int rightIR = digitalRead(R_S);

  if (leftIR == BLACK_VALUE && rightIR == BLACK_VALUE) {
    forward();
    lastTurn = 0;
  }
  else if (leftIR == BLACK_VALUE && rightIR != BLACK_VALUE) {
    strongLeft();
    lastTurn = 1;
  }
  else if (leftIR != BLACK_VALUE && rightIR == BLACK_VALUE) {
    strongRight();
    lastTurn = 2;
  }
  else {
    findLineStrong();
  }
}
```

Obstacle Avoidance and Path Scanning Logic

```
if (frontDistance > 0 && frontDistance <= OBSTACLE_DISTANCE) {
  obstacleAvoid();
  return;
}

void obstacleAvoid() {
  Stop();
  beepTwoTimes();

  scanServo.write(LEFT_ANGLE);
  delay(700);
  int leftDistance = getDistance();

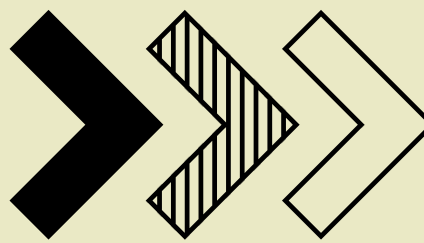
  scanServo.write(RIGHT_ANGLE);
  delay(700);
  int rightDistance = getDistance();

  scanServo.write(CENTER_ANGLE);

  if (leftDistance > rightDistance) {
    avoidLeft();
  } else {
    avoidRight();
  }

  searchLineAfterObstacle();
}
```

CODE SNIPPETS



PIR-Based Danger Detection Logic

```
if (pirState == HIGH) {
    dangerAlert();
    return;
}

void dangerAlert() {
    Stop();

    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("DANGER ALERT!");
    lcd.setCursor(0, 1);
    lcd.print("Motion Detected");

    while (digitalRead(PIR_PIN) == HIGH) {
        Stop();
        digitalWrite(BUZZER_PIN, HIGH);
        delay(250);
        digitalWrite(BUZZER_PIN, LOW);
        delay(200);
    }
}
```

GSM-Based SMS Alert Logic

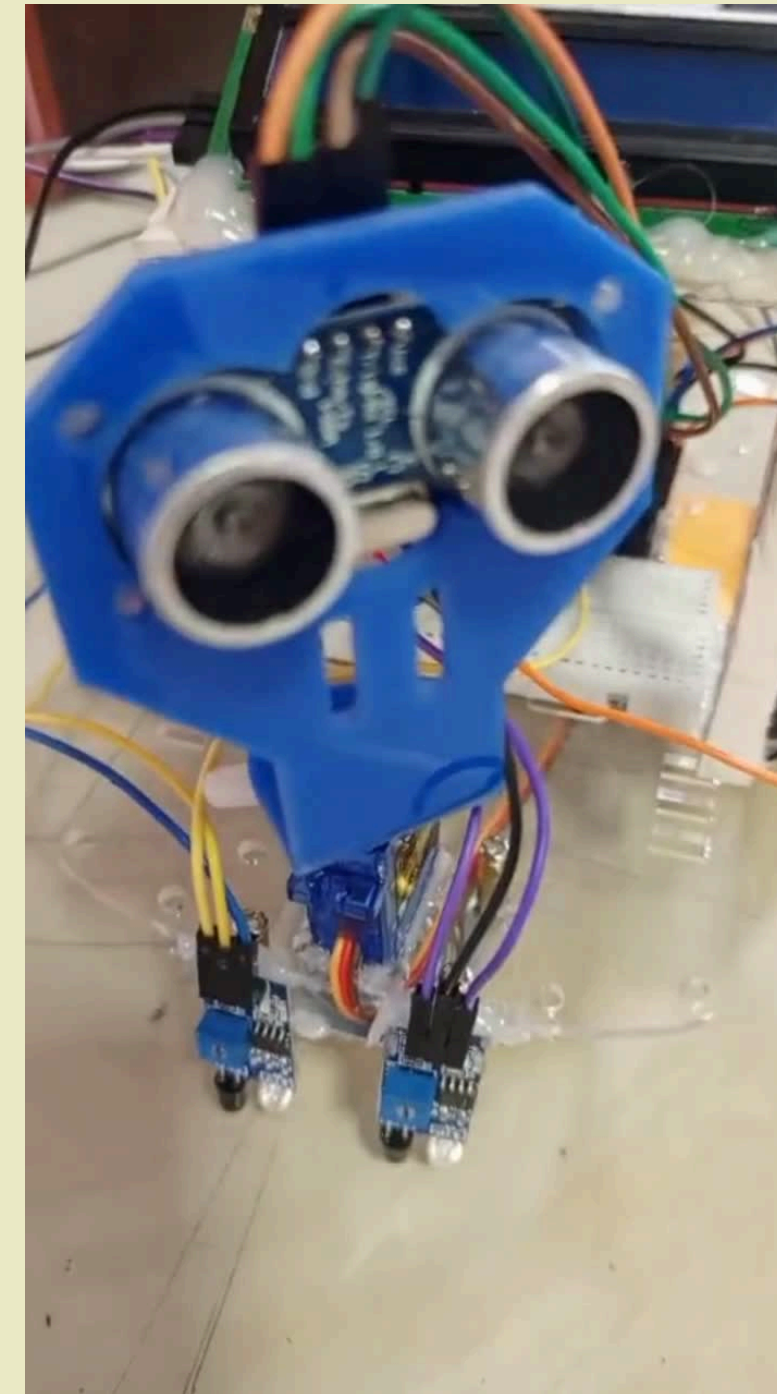
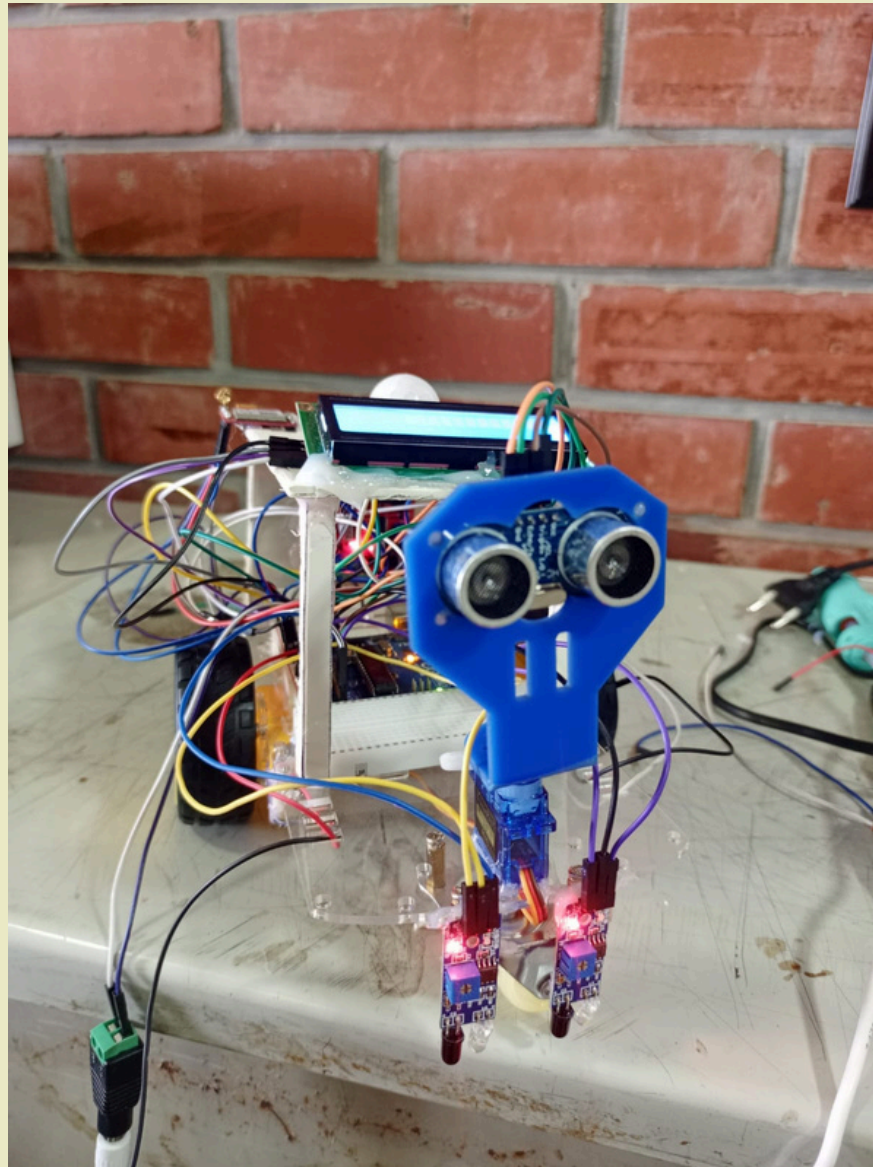
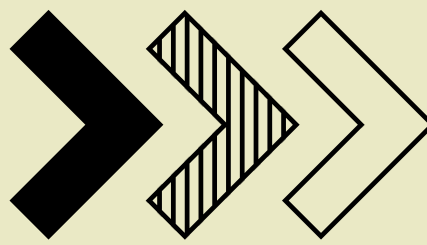
```
void sendSMS(String message) {
    gsm.println("AT+CMGF=1");
    delay(500);

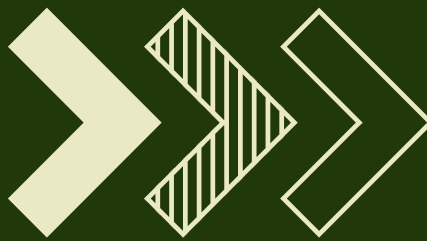
    gsm.print("AT+CMGS=\");
    gsm.print(PHONE_NUMBER);
    gsm.println("\");
    delay(500);

    gsm.print(message);
    delay(500);

    gsm.write(26);
    delay(3000);
}
```


ROBOT DEMO VIDEO





Thank you





Any Question ?